

Blood Water Distillation and CRDS:

- 1.) Obtain ice and materials to conduct the procedure, and allow RBC samples to thaw for use.
- 2.) Place thawed samples in a homogenizer without the use of beating beads. Run for 1 min.
- 3.) Spin down the homogenized RBC in centrifuge at 20,000g for 15 minutes at 4C in order to obtain cytosolic supernatant and cell pellet.
- 4.) Draw 100 uL of cytosolic supernatant and place in the center of an inverted and labeled O-ring cap.
- 5.) Carefully screw on the rest of the Eppendorf tube, and place in heated (80C) sand for overnight (8-10 hours) distillation.
- 6.) Retrieve distillate and carefully replace the old cap with a new labeled cap.
- 7.) Collect distillate through centrifuging sample at 2000g for 2 minutes. If storage is needed, store at -20C.

8.) Place all standards and samples in ice while not being used

9.) Create a 1/300 dilution for all standards and samples

- Add 1500uL fresh ddH₂O to LCMS Vial
- Add 5uL sample/standard to LCMS Vial

Standards and sample dilutions should be run in triplicate as technical replicates.

10.) At LGR facility, place all samples (wrapped with thin layer of parafilm in appropriate testing tray)

11.) Upload Pre-made CSV file for LGR analysis (see CN Program for details)

- 8 injections per sample/standard
- Randomized blocks for samples

12.) At the end of the Run, gather data from CRDS using a USB

LGR (D₂O) Prep & Acquisition (Detailed)

Distillation and Dilution of Sample Water

Sample Prep:

- 1) Obtain ice (H₂O) to chill samples while working.
- 2) Retrieve mouse blood/ plasma from the fridge or freezer.
 - Note: (if from -80 freezer, store on dry ice while sorting samples)
- 3) Allow blood/plasma samples to thaw in (H₂O) ice. Avoid over-thawing samples.

Distillation Prep & Procedures: (Complete these steps in biohazard hood if human samples)

- 4) Label some O-Ring screw cap tube for each of the samples you will be distilling. Place the O-Ring cap upside down on a steady surface.
 - 5) Using a micropipette, place 50-100 µl of the liquid blood/ plasma in the center of the O-ring screw cap.
 - You may need to hand incubate samples just before transferring to the O-ring cap.
 - Place enough blood/plasma to mostly fill the center of the cap. Avoid overfilling or creating cohesive bubble
 - Return remaining blood/plasma to
 - 6) With the tube inverted, carefully screw the cap on (without any sample touching the tube).
 - 7) Place the inverted tube into a sand bath overnight at 80°C for distillation
 - Track actual temperature with a glass thermometer.
 - Set Sand Bath to about 110 to yield an actual 80°C.
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Retrieving Distillate:

Once sand bath distillation is complete you should see a collection of distillate at the tip of the O-ring tube.

- 8) Carefully Remove the distillation tube from the sand bath while keeping inverted..
 - You may need to blow off the tube with light air flow to remove leftover sand.
 - Any sudden motions may cause the distillate to fall back into the cap and contaminate the tube.
- 9) While inverted, remove the tube cap (which contains the distillate) and replace the cap with a new one. Tighten cap on the tube.
- 10) Further collect the condensate by centrifugation (2000 x g for 5 min)

This distillate can be saved at negative 20 for up to a week (if O-ring cap on the tube is tightly sealed).

LGR Sample Prep:

- 11) Calibrate all the pipettes you will be using (p20s,p1000s)
 - Review the section titled “Calibrating Pipettes” (Steps 25-26) for more details
- 12) Obtain H₂O ice to chill LGR Standards while working.
- 13) Obtain fresh 18 Mega Ohm water (ddH₂O) from downstairs MS Room, and cap in a large tube.
- 14) Obtain the LGR Enrichment Standards from the Cell culture room -20C freezer. Allow the LGR standards to thaw in ice until time of use.
 - Note: If more standard volume is needed, or these standards are not available, review section titled “Creating LGR D2O Enrichment Standards” (Steps 22-24).
- 15) Obtain and label small GC vials and teflon caps (VWR p/n: 66016-078) according to the samples and standards needed
 - 1 vial for each standard (7 total)
 - 1 vial for each sample distillate you will analyze
 - Note: prepare sample distillates in groups of 20 or less for an overnight run

Diluting Standards & Samples for LGR Range:

Create a 1:300, (sample/standard: ddH₂O) dilution.

- 16) Using a micropipette, measure out the needed volume of ddH₂O and place into the GC vial. (Volumes shown below)
- 17) Vortex each sample/standard for 3 seconds, measure the needed volume and place in the GC vial to complete dilution. (Volumes shown below)

	0%	0.5%	1%	2%	4%	6%	10%	Sample Dilutions
Sample/ Standard Volume	5 µl	5 µl	5 µl	5 µl	5 µl	5 µl	5 µl	5 µl
ddH ₂ O Volume	1500 µl	1500 µl	1500 µl	1500 µl	1500 µl	1500 µl	1500 µl	1500 µl

- 18) Tighten the caps of all vials to the second level when dilutions are complete.
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Analysis of Water using LGR:

- 19) Got to LGR Machine found in Eyring Building 314

20) Place all samples in one of the trays in an organized manner so that you know where each standard and sample is.

- Wrap each GC vial with parafilm to create thickness around the bottle. This will prevent the robot from pulling the whole vial out of the tray and breaking the glass.

21) Open the LGR Interphase and Set up the Standards

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Creating LGR D₂O Enrichment Standards: (As Needed)

22) Obtain Deuterated Water (found in Human Research Box in the cell culture room. Chain code: 10-24-14.)

23) Collect ddH₂O (18 MOhm) from downstairs in a large eppendorf tube.

24) Pipette a set volume of D₂O and add additional ddH₂O to a total volume needed. (standards shown below.)

Standards:

	0	.5	1	2	4	6	10
H ₂ O	100%	99.5%	99%	98%	96%	94%	90%
D ₂ O	0%	0.5%	1%	2%	4%	6%	10%

Calibrating Pipettes:

25) Obtain the plastic wrench from the upper drawer on the north side of the sink.

26) Check the actual accuracy of the pipette using the quantitative scale

- Tare the scale and a weigh boat. Then add a specific volume of H₂O as your theoretical delivery volume.
- Using the density of pure water (1g/mL), calculate the actual volume delivered and compare it to the theoretical volume that it should have delivered.
- To adjust the volume, insert the plastic wrench into the calibration hole on the front side of the pipette, and turn"
 - Clockwise (CW) to increase delivery volume
 - Counter-Clockwise (CCW) to decrease delivery volume
- **NOTE:** avoid over turning wrench as calibration may be less accurate as the calibration wears down.

Analysis of Water using LGR (bring syringe if applicable)

27) To run samples on the instrument in the Eyeing building with the standards, choose 8 injections per sample/standard and that you choose at least 3 replicate

runs. Make sure that the initial injections (first several should say “norm”) are running correctly before leaving the instrument.

- 28) Place samples in trays (be organized about it so we know which positions are which samples. We do this because you will program the computer to pick certain positions and those positions will be named by the sample!). Wrap each vial with enough parafilm so that the sampling needle CANNOT pull vial out of the tray!
- 29) Select the Configure button at the bottom of the screen.

